Karan Kapur

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Education

University of Wisconsin-Madison — Madison, WI

Bachelor of Science in Computer Science and Data Science, Minor in Entrepreneurship — *May 2026*

- Relevant coursework: **Artificial Intelligence**, **Big Data Systems**, Operating Systems, Algorithms

Experience

Data Analyst — 08/2026 - Present

Wisconsin Union — *Madison, WI*

- **Replaced the team's old scheduling software with an app I shipped in 2 days, used daily by 250+ employees**: 4,400+ shifts, self-serve accounts, and work chat; built on TypeScript, MySQL, Docker, and AWS.
- **Forecasted weekly food demand and staffing** from sales and shift data in Excel to guide orders and schedules.

Co-Founder & Cloud Storage Engineer — 04/2026 - Present

thaw — *Madison, WI*

- **Cut LLM session cold boots on H100s ∼386x, from ∼340s to a 0.88s median fork**: co-founded an open-source inference startup (Rust, CUDA) with N. Matteson and M. Yu.

Software Engineer — 01/2026 - 05/2026

Eurofins Scientific — *Madison, WI*

- **Shipped a forecasting product lab technicians used daily**: partnered with Eurofins as a client through UW-Madison, interviewed technicians and chemists across **4+ business units**, and took it to **production**.
- **Turned 10K+ messy records into forecasts with confidence scores**: built the ETL pipelines, **PostgreSQL** schema, **FastAPI** backend, and **React** front end; ran sprints as Scrum Master.

Software Engineer — 06/2025 - 08/2025

OpenAI x UW-Madison SAIL Program — *Madison, WI*

- **Reduced manual lookup time 40% and cloud cost 25% for 60+ users in UW-Madison IT**: built a GPT-4 **RAG** platform serving 50+ concurrent queries at sub-2s latency.

Computer Support Assistant — 01/2025 - 05/2026

UW-Madison, Department of Electrical & Computer Engineering — *Madison, WI*

- **Resolved 1,000+ time-sensitive issues at 98% satisfaction**; monitoring automation lowered downtime 40%.

Automation & Compliance Engineering Intern — 07/2024 - 09/2024

HCLSoftware — *Noida, India*

- **Lifted onboarding conversion 30%** via an analytics-driven redesign; 50+ tests shortened regression 60%.

Projects

tell (solo build) | Explainable buying-signal pipeline | github.com/karank2512/tell-ai — 10/2026

- **Shrank entity-resolution review load 85% (625 to 91) on 16,243 held-out SEC Form D filings**: guarded name links by issuer type, incorporation date, and fund status.
- **Deployed 6 collectors on a 4x/day GitHub Actions cron at $0/month**: job-board APIs, SEC EDGAR, and news into Neon Postgres; sha256 event IDs make reruns insert 0 rows, and CI replays every collector twice.

Foreman (solo build) | **Agentic** workforce platform | github.com/karank2512/Foreman — 09/2026

- **Lowered cost per run 30% vs an all-LLM pipeline on a 100-record paired benchmark**: routed 5 of 7 steps to deterministic code, taking LLM calls per run from 11 to 6 (list prices).
- **Prevented repeated side effects in all 10 tested crash and race failure modes**: a Postgres-backed run queue with leases, heartbeats, crash recovery, and idempotent resume restores a crashed run within 3 minutes.

RelayIQ (solo build) | **Agentic** decision engine | github.com/karank2512/RelayIQ — 07/2026

- **Raised precision from 0.682 to 0.773 across six strategies**: calibrated confidence (Brier 0.164, ECE 0.091) holds low-confidence writes back from **HubSpot** and escalates them to a human in real time.

Technical Skills

Languages and Stack: **Python**, **TypeScript**, **SQL**, Rust, C++, Next.js, React, FastAPI, PostgreSQL, Docker, AWS
AI and Agents: LLM tool calling, **agents**, RAG, LLM-judge **evals**, calibration, Vercel AI SDK, MCP, vLLM, CUDA
GTM and Tools: **HubSpot** APIs, webhooks, Clay and Snowflake (familiar), **Claude Code** daily, Cursor